

Understanding the Laws of Motion: A Beginner's Guide

Lesson 6: Free Body Diagrams

Free Body Diagrams - Study Notes

Key Concepts

- **Isolation:** The process of focusing solely on the object you're analyzing and ignoring everything else in the environment.
- **External Forces:** Forces acting *on* the object from outside sources (not forces *within* the object itself).
- **Vectors:** Quantities with both magnitude (size) and direction. Forces are always represented as vectors.
- **Newton's Second Law Connection:** Free body diagrams are the foundation for correctly applying Newton's Second Law ($F = ma$).
- **Representation:** Using arrows to visually depict forces, where arrow length indicates magnitude and arrow direction indicates direction.

Summary

A free body diagram (FBD) is a fundamental tool in physics used to analyze the forces acting on a single object. The core idea is to isolate the object of interest and represent all external forces acting *on* it as vectors. This means drawing arrows where the length of the arrow corresponds to the force's magnitude (strength) and the arrow's direction indicates the force's direction. It's crucial to only include forces acting *on* the object, not forces the object exerts on other things.

Creating an accurate FBD is the first and most important step in solving many physics problems involving forces. Without a correct FBD, applying Newton's Second Law ($F = ma$) will likely lead to incorrect results. By visually representing the forces, you can clearly see how they combine and determine the net force acting on the object, which is essential for calculating its acceleration.

Important Points

- **Always isolate the object:** Clearly define what object you are analyzing.
- **Only include *external* forces:** Don't include internal forces (like the force of a spring within the object itself).
- **Represent forces as vectors:** Use arrows to show both magnitude and direction.
- **Label each force:** Clearly label each force (e.g., Gravity, Normal Force, Applied Force, Tension).
- **Common Forces to Consider:**
 - * **Gravity (F_g):** Always acts downwards towards the center of the Earth.
 - * **Normal Force (F_n):** A force exerted by a surface perpendicular to the surface.
 - * **Applied Force (F_a):** A force directly applied to the object (e.g., a push or pull).
 - * **Tension (T):** The force transmitted through a string, rope, cable, or wire.
 - * **Friction (F_f):** A force that opposes motion between two surfaces in contact.

- **Direction is Key:** Pay close attention to the direction of each force. Incorrect direction = incorrect answer.
- **Coordinate System:** Consider establishing a coordinate system (x and y axes) to help resolve forces into components if needed.

Examples

Example 1: Block on a Table

- **Object:** A block resting on a table.
- **Forces:**
 - * **Gravity (F_g):** Pointing downwards. Magnitude is mg (mass * acceleration due to gravity).
 - * **Normal Force (F_n):** Pointing upwards, exerted by the table. Magnitude is equal to the force of gravity (in this case, since the block isn't accelerating vertically).
 - **FBD:** A simple diagram showing a block with a downward arrow labeled " F_g " and an upward arrow labeled " F_n ". The arrows should be roughly equal in length.

Example 2: Box Being Pulled Across a Floor

- **Object:** A box being pulled across a floor by a rope.
- **Forces:**
 - * **Gravity (F_g):** Pointing downwards. Magnitude is mg .
 - * **Normal Force (F_n):** Pointing upwards, exerted by the floor. Magnitude is equal to the force of gravity.
 - * **Applied Force (F_a):** Pointing horizontally in the direction of the rope's pull.
 - * **Friction (F_f):** Pointing horizontally, opposing the motion of the box.
 - **FBD:** A diagram showing a box with a downward arrow labeled " F_g ", an upward arrow labeled " F_n ", a horizontal arrow labeled " F_a " pointing right, and a horizontal arrow labeled " F_f " pointing left.

Quick Review

1. What is the primary purpose of a free body diagram?
2. Why is it important to only include *external* forces in a free body diagram?
3. In a free body diagram, what does the length of an arrow represent?
4. Give an example of a common force you would include in a free body diagram.
5. True or False: Internal forces are important to include in a free body diagram.

Further Reading

- **Newton's Laws of Motion:** A deeper understanding of Newton's Laws is crucial for using FBDs effectively.
- **Vector Addition:** Learning how to add vectors is essential for calculating the net force.
- **Friction:** Explore the different types of friction and how to calculate frictional forces.
- **Inclined Planes:** Analyzing forces on objects on inclined planes is a common application of FBDs.
- **Tension in Ropes:** Understanding how tension forces work in ropes and pulleys.

